

# Resilience performance analysis of interdependent infrastructures using Fault Tree and Dynamic Bayesian Network

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## ABSTRACT

Interdependent infrastructure systems are designed to work synergistically, but disruptions in one can rapidly propagate to others, posing significant challenges to their stable operation. This study introduces a novel integrated probabilistic modelling framework for simulating the resilience performance of interdependent infrastructures. The proposed computational model employs a Fault Tree to interpret complex relationships among interdependent infrastructures, identify cascading failure propagation paths, and calculate component failure probabilities in non-disaster scenarios. Building on the static cascading failure analysis provided by the fault tree, a hybrid Dynamic Bayesian Networks model is developed to assess the dynamic resilience of interdependent infrastructures, incorporating both constant nodes (unaffected by disasters) and time-dependent nodes (affected by disasters). The proposed computational model is applied to a real community-scale water–electricity–road (WER) system in China, with a focus on physical interdependencies. Under non-disaster conditions, the probability of water-supply failure (1.89%) exceeds that of electricity supply (0.67%) and road accessibility (0.58%), with ageing sub-pipelines identified as critical vulnerable components. Under rainfall scenarios, resilience performance loss increases nonlinearly with rainfall intensity, with the road system exhibiting the most rapid degradation, followed by the electricity and water systems. These results support a sequenced recovery and mitigation strategy that prioritises roads, followed by electricity, and then water. Overall, this study provides a novel integrated probabilistic model to identify critical physical interdependencies, assess time-iterative resilience performance, and compare recovery strategies, thereby supporting resilience-informed decision-making for disaster mitigation and protection of critical infrastructures.

## 1. Introduction

Infrastructure networks are the arteries of modern cities, as their construction and function significantly impact security, economy, and other critical domains [1]. These systems, each serving distinct functions, exhibit complex interdependencies, particularly those that support urban essential life, such as electricity, water, roads, and communication networks. For example, water supply systems rely on electricity for pump operation, while road networks are crucial for the maintenance and repair of electrical infrastructure. The interdependency between infrastructure systems has two sides. On the one hand, during disaster events, these interdependencies can trigger cascading failures among infrastructure systems, amplifying the impact of the

disaster. On the other hand, understanding and mapping these interdependencies is beneficial for setting protection and response strategies and enhancing coordination among different institutions contributing to infrastructure disaster mitigation [2,3].

The increasing interdependency of infrastructure networks heightens their vulnerability. Even a minor disruption in one network can cascade to affect other dependent networks, resulting in significant adverse impacts [4]. Over the past two decades, the intensification of global warming has led to more frequent extreme weather events and significant natural disasters [5]. Floods, bushfires, hurricanes, and heatwaves have repeatedly disrupted power grids, transportation systems, and communication networks, triggering cascading failures. For example, thunderstorms struck Victoria on February 13, 2024, damaging the

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electrical infrastructure and knocking out electricity for about 130,000 people [6]. The power outage led to a series of cascading failures, including lack of electricity triggered the loss of network signals, preventing users in the affected areas from communicating with the outside world. In September 2017, Hurricane Maria struck Puerto Rico, causing extensive damage to the electrical, water, and transportation systems [7]. The collapse of the electrical system led to communication disruptions, healthcare disruptions, and other issues that exacerbated the impact of the disaster. Such frequent events arising from climate change pose substantial threats to urban infrastructure networks, resulting in frequent and severe losses, underscoring the urgent need for resilience performance analysis for interdependent infrastructures [8–10]. Here, resilience performance refers to the ability of interdependent infrastructure systems to maintain or quickly restore essential services when disruptions propagate across sectors, typically characterised by the magnitude of service degradation and the speed of recovery [11]. Analysing resilience performance of interdependent infrastructures provides a more realistic basis for disaster reduction and recovery strategy through a multisystem perspective, helping decision-makers prioritise what to protect, what to restore first, and which strategies can most effectively accelerate system-level service recovery.

Existing resilience-performance studies have made substantial progress in modelling disruption–propagation–recovery processes for interdependent infrastructures, however, three limitations remain that directly motivate this research. First, a large portion of the literature adopts network-based cascading simulations that are effective for exploring propagation and recovery heuristics [12,13], yet they often represent interdependencies using simplified coupling rules and lack an explicit representation of physical interdependencies. Second, several decision-oriented models often emphasise optimal recovery under constraints [14,15], their outputs are typically system-level, which constrains the assessment and comparison of component-level vulnerability (e.g., which components should be prioritised for protection and which fail first under uncertainty). Third, although probabilistic approaches such as BBNs [16] have been applied, few studies have systematically linked static dependency logic and non-disaster component failure probability to time-iterative resilience performance for interdependent infrastructures.

Compared with prior resilience studies that rely primarily on network-based simulations or optimisation-only restoration models, this study proposes a unified probabilistic, time-iterative computational model for interdependent infrastructures. Specifically, (i) the model employs a Fault Tree to formalise dependency logic and quantify component failure probabilities under non-disaster conditions, (ii) it embeds this information within a Dynamic Bayesian network model to simulate resilience performance together with hazard-dependent degradation and recovery over discrete time steps, and (iii) this model outputs comparable component-level failure probabilities and system-level resilience performance trajectories to support prioritisation of protection and recovery strategies in water-electricity-road system.

The remainder of this paper is organised as follows: Section 2 reviews the literature on resilience performance in interdependent infrastructure systems. Section 3 details the proposed resilience performance assessment and simulation methodology. Section 4 presents a case study on an interdependent water distribution, electricity distribution, and road network in a community in China using the proposed methodology. Section 5 presents the resilience performance simulation results. Section 6 conducts the comparative analysis of different disaster strengths and recovery strategies. Finally, Section 7 summarises the key findings, discusses research limitations, and suggests future research directions.

## 2. Literature review

Current research on the resilience performance of interdependent infrastructure primarily focuses on how the interdependency between multiple systems influences overall resilience. Studies have shown that

evaluating the resilience of a single infrastructure system is insufficient to reflect resilience performance because disruptions can propagate across sectors [17–21]. Rinaldi, Peerenboom [22] et al. classify critical-infrastructure interdependencies into four categories (physical, cyber, geographic, and logical) to capture service coupling, information control reliance, location exposure, and rule-driven dependencies. However, cascading failures and interdependencies describe related but distinct mechanisms. Cascading failures refer to the progressive propagation of failures initiated by an initial disruption, often studied as intra-network dynamics. By contrast, interdependencies describe inter-network couplings in which the operation of one infrastructure relies on the state or service of another. When multiple infrastructures are considered as an integrated system-of-systems, interdependencies act as cross-sector pathways that can transmit disruptions between networks, thereby producing multi-system cascading failures. Accordingly, this study explicitly models cross-infrastructure interdependencies and simulates time-dependent resilience performance (cascading failures propagation and recovery) across multiple infrastructure systems.

Based on the studies summarised in Table 1, current research on resilience performance of interdependent infrastructures typically models (i) interdependency structures between sectors and (ii) the disruption–propagation–recovery process under hazard or disruption scenarios. Common inputs include network topology, component attributes, disruptions, and recovery resources or rules. Corresponding outputs are usually time-varying performance trajectories and resilience indicators, such as service availability over time, recovery time, network efficiency or connectivity, expected unmet demand not supplied, and area-based resilience metrics, often accompanied by identification of critical components and comparison of alternative recovery strategies.

Representative mainstream approaches for characterisation of interdependent infrastructure resilience performance include network theory [29–31], Bayesian Networks [32–35], and Agent-based modelling [36–39]. Network theory methods emphasise network topology but struggle to capture the dynamics of failure propagation. Agent-based modelling effectively simulates interactions and autonomous decisions within infrastructure networks, while exhibiting limitations in quantitative probabilistic prediction. Traditional Bayesian networks represent static causal relationships among failures yet lack the capability to handle dynamic temporal aspects.

However, Dynamic Bayesian Networks (DBNs) are particularly well-suited for evaluating resilience performance in interdependent infrastructure systems due to their ability to model temporal and probabilistic dynamics. Unlike static models, DBNs capture how system states and interdependencies evolve over time, providing a realistic representation of cascading failures and recovery processes. Their strength lies in probabilistic forecasting under uncertainty, which is essential for anticipating disruptions and assessing system resilience performance [40]. In addition, the resilience performance is not just related to disaster strikes; the ageing and usage status of infrastructure also need to be considered. Fault Tree Analysis (FTA) is a widely adopted technique in engineering fields for identifying potential causes of system failures in operation status, particularly valued for its structured and systematic approach. As outlined by Baig et al. (2020), FTA provides a clear, top-down visualisation of failure pathways, facilitating comprehensive failure causes in complex systems.

To simultaneously leverage the advantages of static causal analysis and dynamic prediction, this research integrates Fault Tree (FT) and Dynamic Bayesian Networks (DBNs) methodologies. Specifically, FT is first utilised to construct logical structures of infrastructure failures, which are then transformed into DBNs model for dynamic, probabilistic, and temporal forecasting. Compared to other methods, the DBNs approach excels in capturing the temporal evolution and uncertainty of infrastructure states, providing superior probabilistic predictions of cascading failure processes [40–43]. The combined FT-DBN method [44] enables accurate quantitative assessments of failure probabilities

**Table 1**

Reference comparisons for interdependent infrastructure resilience performance.

| No. | Reference                    | Application domain                                                                  | Hazard or disruption                      | Interdependency types considered                                     | Period covered                                                | Approach streams                                            | Specific approach                                                                                                                                               | Resilience performance (outputs)                                                                                     | Main findings                                                                                                                                                                     |
|-----|------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------|----------------------------------------------------------------------|---------------------------------------------------------------|-------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1   | Nasrazadani and Mahsuli [13] | Community infrastructures (power, water, cellular) with transport influence         | Hurricane Ike (2008) case                 | Physical; Logical; Geographic (common exposure)                      | During & post                                                 | Probabilistic modelling                                     | Dynamic Integrated Network (DIN II): dynamic inoperability & recovery with multi-level dependencies and uncertainty                                             | Inoperability trajectories, time to recovery                                                                         | Ignoring system-to-facility dependencies can significantly underestimate recovery time                                                                                            |
| 2   | Kong, Zhang [12]             | Regional multi-energy (electricity–gas–oil), Greater Toronto Area                   | Regional natural hazards (scenario-based) | Physical; Geographic (proximity-based); Logical (budget constraints) | Planning & during & post                                      | Optimisation & simulation (network-based)                   | Genetic algorithm & Monte Carlo simulation to select combined resilience improvement measures under constraints                                                 | Expected resilience under constraints                                                                                | Optimal mix of measures outperforms single-measure upgrades; shows benefits of combined interventions                                                                             |
| 3   | Nan and Sansavini [23]       | Interdependent infrastructures; electric power supply system (EPSS) demonstration   | Disruptive events (scenario-based)        | Physical; Cyber; Logical                                             | During & post (explicit phases)                               | Hybrid simulation (agent-based & power-flow)                | Hybrid multi-layer modelling (physical & SCADA/control & human operators); ABM with DC power flow & human reliability concepts                                  | Resilience metric across phases; recovery trajectories                                                               | Highlights the influence of control and human layers on resilience outcomes; supports phase-based resilience evaluation                                                           |
| 4   | Kanti Sen, Dutta [16]        | Interdependent buildings–roadway system (community)                                 | Flood                                     | Geographic; Physical                                                 | During & post                                                 | Probabilistic graphical model (BBN) & GIS                   | GIS-based preprocessing & Bayesian Belief Network (BBN); sensitivity analysis for critical components                                                           | Probabilistic resilience metric values                                                                               | Road damage affects recovery; sensitivity identifies critical nodes/components for decision support                                                                               |
| 5   | Chopra, Dillon [24]          | Urban electricity distribution & metro & repair resource networks (London)          | Flooding (hypothetical local event)       | Physical; Logical (resource utilisation)                             | During & post (24 h simulation)                               | Network flow & optimisation (resource-constrained recovery) | Dynamic network flow (min-cost) & repairable-systems model & rolling-horizon LP for repair resource allocation; scenario tree for uncertainty                   | Time-varying operability; recovery time                                                                              | Rolling-horizon repair allocation improves recovery; captures multiple dependency types, including repair resources                                                               |
| 6   | Wang, Gu [25]                | Interdependent power–gas system (artificial network)                                | Abstracted attacks                        | Physical (coupling modes)                                            | During & post                                                 | Network-based cascade simulation & deep learning            | CNN topology identification (adjacency-matrix classification) & cascade simulation & resilience triangle; compare recovery strategies                           | Recovery strategy performance                                                                                        | Coupling modes and attack strategies strongly affect resilience; proposed recovery strategy shows best performance.                                                               |
| 7   | Yodo and Arfin [26]          | Interdependent gas–power system with microgrids (IEEE 14-bus & 20-node gas network) | Abstract disruptions                      | Physical; Logical (budget/time constraints)                          | During & post (& planning via microgrid deployment decisions) | Network-based discrete-time simulation                      | Discrete-time cascade & recovery; network efficiency performance metric; optional optimisation under time                                                       | Network efficiency over time                                                                                         | Microgrids substantially improve resilience; optimisation supports determining required microgrid quantity                                                                        |
| 8   | Cardoni, Cimellaro [27]      | Buildings–electrical distribution system for seismic resilience                     | Earthquake                                | Geographic; Physical                                                 | During & post                                                 | Physics-based simulation                                    | Fragility-based building damage assessment; rule-based building to substation failure coupling; power-system analysis using MATPOWER; spatial network synthesis | Proposed resilience index value                                                                                      | Hosting-building damage can drive substation failure and materially change resilience results; the proposed index reflects redundancy and resourcefulness                         |
| 9   | Chen, Wang [28]              | Interdependent public transport (bus–metro) network (Wuhan)                         | Abstracted attacks                        | Geographic (distance-based coupling); Physical (transfer coupling)   | During & post                                                 | Network theory & load–capacity cascade simulation           | Two-layer coupling by distance & load capacity cascade & five recovery strategies                                                                               | Structural & functional resilience; recovery comparison                                                              | Coupling distance and recovery strategy strongly influences resilience; compare strategies for faster functional restoration                                                      |
| 10  | <b>This study</b>            | <b>Community water–electricity–road system</b>                                      | <b>Heavy rain</b>                         | <b>Physical</b>                                                      | <b>non-disaster &amp; during &amp; post</b>                   | <b>Probabilistic temporal model</b>                         | <b>Fault Tree Analysis for baseline failure logic &amp; Dynamic Bayesian Network for time-dependent degradation and recovery simulation</b>                     | <b>Both Component and system failure probabilities, resilience performance trajectories, and strategy comparison</b> | <b>Identifies vulnerable components under normal and rainfall scenarios; compares rainfall intensities and recovery sequencing to improve multiple infrastructures resilience</b> |

and trends.

According to the literature review, we identify that a key gap is the limited availability of a computational model that simultaneously: (i) represents explicit physical interdependencies among infrastructure sectors rather than simplified coupling rules), (ii) produces component-level probabilistic failure estimates to support prioritisation under uncertainty, and (iii) links static dependency logic and non-disaster failure probabilities to time-iterative resilience performance trajectories for interdependent infrastructures.

To demonstrate the proposed computational model, the study presented in this paper chooses three infrastructure systems: water supply, electricity supply, and road network, due to their strong interdependencies as well as their key role in ensuring the basic normal life of the population after the unexpected disturbance. The proposed computational model employs the FT model to interpret complex relationships among interdependent infrastructures, identify cascading failure propagation paths, and quantify the probability of failure events considering component fatigue and ageing. Building on the static cascading failure analysis provided by the FT model, a hybrid DBNs model is developed. This model incorporates both constant nodes (unaffected by disasters) and time-dependent nodes (affected by disasters) to simulate the temporal dynamics of resilience performance in IIRs.

### 3. Methodology for modelling resilience performance

In this study, a methodology integrating Fault Tree (FT) analysis and Dynamic Bayesian Networks (DBNs) is proposed for resilience performance analysis. The resilience performance evaluation is based on resilience performance curve of critical components. This methodology can be applied to model various disaster scenarios. To simulate resilience performance, this paper constructs degradation and recovery models for critical components. The proposed methodology for resilience performance analysis is illustrated in Fig. 1.

This methodology consists of 3 main steps: (1) characterisation of IIRs and disaster effect, (2) identifying the common causes of performance failure of IIRs in non-disaster scenario, and (3) modelling resilience performance under disaster scenarios. Step 1 involves identifying the critical components of IIRs, defining the correlations between these components, and developing the theoretical model of disasters' impacts on components. In step 2, the potential causes of performance failure in

the IIRs system are comprehensively identified non-disaster scenarios using FT model. In step 3, based on the failure causes identified by the FT, the failure probability for critical components is calculated under disaster scenarios after determining the specific effects of external disturbances. Finally, the corresponding DBN model is constructed to simulate the degradation and recovery process, analysing the temporal dynamics of resilience performance.

#### 3.1. Characterisation of IIRs and disaster effects

This study focuses on analysing resilience performance at the components level, such as water pumps, electricity transformers, etc. Identifying critical components and interdependency relationships is essential for hierarchical IIRs. Additionally, modelling the effects of disasters on components is crucial for evaluating resilience performance.

Identifying and abstracting critical components and interdependency relationships is the foundation in analysing hierarchical interdependent infrastructure systems. This involves recognising functionally important nodes and links within each subsystem, uncovering the cross-sector relationships. Disasters cause significant changes in the external environment of critical components, leading to an increased failure probability. The impact of external disasters on critical components is influenced by both the intrinsic parameters of the components and the characteristics of the disaster, which are determined by specific factors. When the system is attacked by external disasters, the resilience performance of the water supply, electricity supply, and road network (WER) system decreases rapidly. Heavy rainfall is selected as the representative disaster scenario for illustration.

The failure probability of equipment, such as water pumps, transformers, and traffic lights, is calculated using the following functions. Eq. (1) defines the hazard-induced failure probability of rainfall exposed equipment under rainfall conditions as the combined contribution of two dominant stressors, high humidity and water submersion:

$$\lambda_{f\_equipment} = k_1 * RH^n + k_2 * D^m \quad (1)$$

Where  $\lambda_{f\_equipment}$  is the failure rate of critical components triggered by high humidity or submersion, where RH is relative humidity, D is the submersion water level,  $k_1$  and  $k_2$  are weighting coefficients, n and m are exponents representing sensitivity depending on equipment type and exposure level. This failure probability model provides the key link

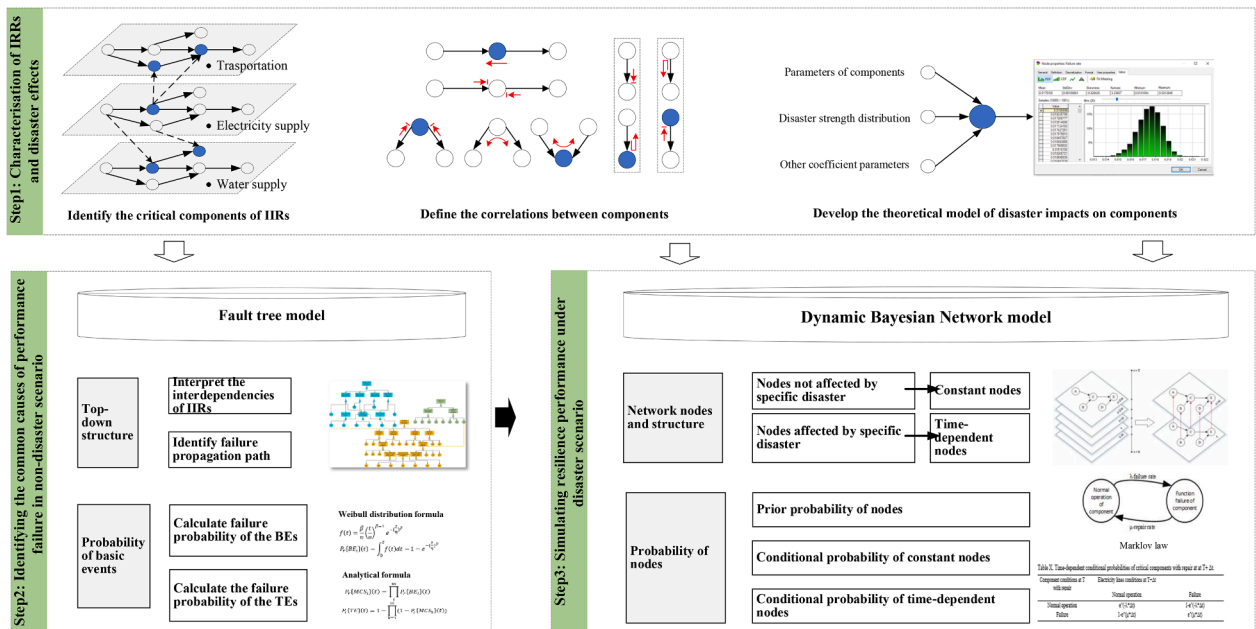


Fig. 1. Flowchart of the proposed methodology.

between hazard intensity and time-dependent component conditions.

Eq. (2) is adopted from Douglas, Garvin [45] specifies rainfall-induced degradation of road accessibility as a function of rainfall intensity.

$$\lambda_{f_{roads}} = k_3 * RI^a \quad (2)$$

Where,  $\lambda_{f_{roads}}$  is the low accessibility rate of roads triggered by rainfall,  $k_3$  is the empirical coefficient of the road segments,  $RI$  is rainfall intensity (mm/h) [46].

### 3.2. Identifying the common causes of performance failure in non-disaster scenario

The FT model provides a structured, top-down approach to systematically identify all common failure causes in interdependent infrastructure systems. Additionally, its graphical representation makes it easy to interpret complex failure relationships and the propagation of failures within interdependent infrastructures.

The FT structure is based on Boolean logic, representing the logical relationships between different faults and their causes [46]. The construction of the FT begins with the top events (TEs) at the root of the tree, progressively breaking it down into its intermediate events (IEs) until the basic events (BEs) are identified. Beyond the FT structure itself, qualitative analysis includes determining the minimal cut sets (MCSs), which represent the smallest collection of basic events that can lead to system failure. The quantitative analysis focuses on evaluating the probability of occurrence for BEs, MCSs, and TEs. Various methods can be used for this analysis, including Monte Carlo simulation, binary decision diagrams, and analytical methods. We adopt the analytical method, which utilises mathematical formulas to approximate the failure probabilities of basic events.

In this study, BEs primarily represent the mechanical failures of critical components, which are susceptible to fatigue and ageing. Since the Weibull distribution is one of the most commonly used life distributions for modelling the failure behaviours of mechanical components, we adopt it to define the failure behaviours of mechanical components. The Probability Density Function (PDF) is calculated below.

$$f(t) = \frac{\beta}{\eta} \left(\frac{t}{\eta}\right)^{\beta-1} e^{-\left(\frac{t}{\eta}\right)^\beta} \quad (3)$$

Eq. (3) characterises the distribution of component failure by component service age, enabling the calculation of the cumulative probability that a component fails by operating time  $t$ . Where  $f(t)$  represents the probability density,  $\beta$  is the shape parameter that determines the failure rate, and  $\eta$  is the scale parameter used to represent the service lifetime.

The occurrence probability of BE ( $P_r\{BE_i\}(t)$ ) with Weibull distribution can be calculated as:

$$P_r\{BE_i\}(t) = \int_0^t f(t)dt = 1 - e^{-\left(\frac{t}{\eta}\right)^\beta} \quad (4)$$

Eq. (4) provides the occurrence probability of each basic event by operating time  $t$ , which is treated as the non-disaster failure probability for the corresponding components.

As the MCSs are presented as the intersection of the statistically independent BEs, the probability of MCS is calculated as:

$$P_r\{MCS_i\}(t) = \prod_{i=1}^m P_r\{BE_i\}(t) \quad (5)$$

MCS probabilities (Eq. (5)) quantify the most critical combinations of component failures that can trigger system-level service loss, thereby supporting identification of dominant failure pathways and vulnerable component sets.

Where  $m$  is the number of basic events in the MCS,  $P_r\{MCS_i\}(t)$  is the occurrence probability of the MCS<sub>*i*</sub> at operating time  $t$ .

Finally, as the TE is the union of the MCSs, the TE probability is approximated as:

$$P_r\{TE\}(t) = 1 - \prod_{k=1}^m (1 - P_r\{MCS_k\}(t)) \quad (6)$$

Where  $P_r\{TE\}$  is the TE's occurrence probability. This TE probability (Eq. (6)) is subsequently used to initialise and calibrate the DBN (e.g., as prior probabilities for constant nodes or initial states for time-dependent nodes), ensuring consistency between the static dependency logic and the dynamic resilience assessment.

### 3.3. Simulating resilience performance under disaster scenario

The FT model is utilised to identify common causes of performance failures and failure propagation paths in IIRs under normal conditions, without accounting for disaster scenarios. Building on this analysis, DBNs are then applied to simulate the resilience performance of IIRs under disaster scenarios. The simulation steps are described below.

#### (a) Construct corresponding DBNs based on FT

In disaster scenarios, the resilience performance of IIRs evolves over time. DBNs provide a compact representation of dependencies across time slices (Fig. 2), effectively simulating this temporal dynamic process. Additionally, DBNs offer powerful inference mechanisms that predict future states based on current and past observations, making them essential for accurately simulating the resilience performance of IIRs.

The nodes and structure of the FT are mapped into the DBNs, preserving their relationships. The event probabilities from the FT are used as prior probabilities for the root nodes in the DBNs, representing the initial conditions of critical components in IIRs. It is worth noting that the basic events that can be triggered by disasters are transferred to time-dependent nodes of DBNs. The basic events that cannot be affected by disasters are transferred to constant nodes of DBNs.

#### (b) Construct the degradation and recovery model of resilience performance

After determining the DBN structure, time-dependent nodes, constant nodes, and prior probabilities from the FT analysis, the next step is to determine the conditional probabilities. For the normal conditional probability, the noisy OR gate and the noisy AND gate are used. Here,  $X_j$  denotes the state of the  $j$ -th parent node and  $Y$  denotes the child node;  $p_j$  represents the conditional transmission probability from  $X_j$  to  $Y$  when  $X_j$  is in the failure state. For the OR gate, the conditional probability is expressed by Eq. (7).

$$P(Y|X_1, X_2, \dots, X_n) = 1 - \prod_{1 \leq j \leq n} (1 - p_j) \quad (7)$$

Eq. (7) specifies situations where the failure of any parent nodes can cause  $Y$  to fail, and the product term represents the probability that none of the active causes successfully triggers  $Y$ . For the AND gate, the conditional probability is expressed by Eq. (8):

$$P(Y|X_1, X_2, \dots, X_n) = \prod_{1 \leq j \leq n} p_j \quad (8)$$

Eq. (8) represents cases where  $Y$  fails only when all required parent conditions are satisfied, Fig. 3 and the product form captures the joint requirement under independency assumptions.

For the time-dependent nodes, the degradation and recovery model is constructed to obey the Markov law (Fig. 4). Specifically, the failure

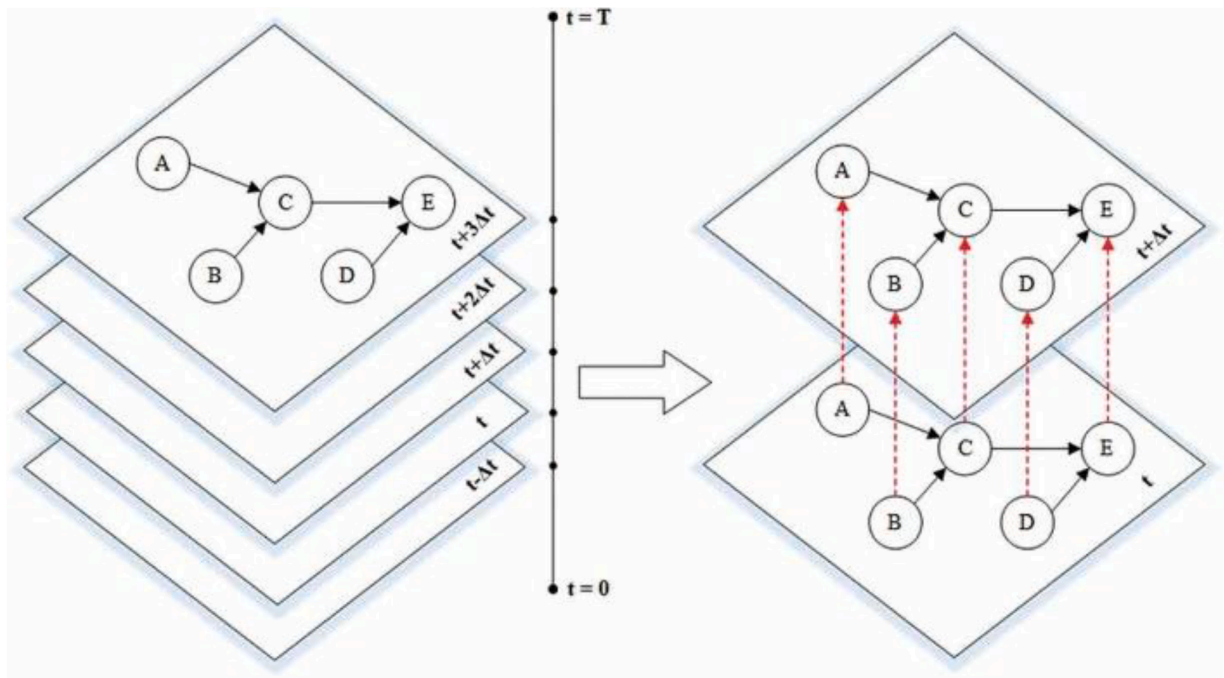


Fig. 2. The DBNs scheme over a discretised timeline  $t$  [42].

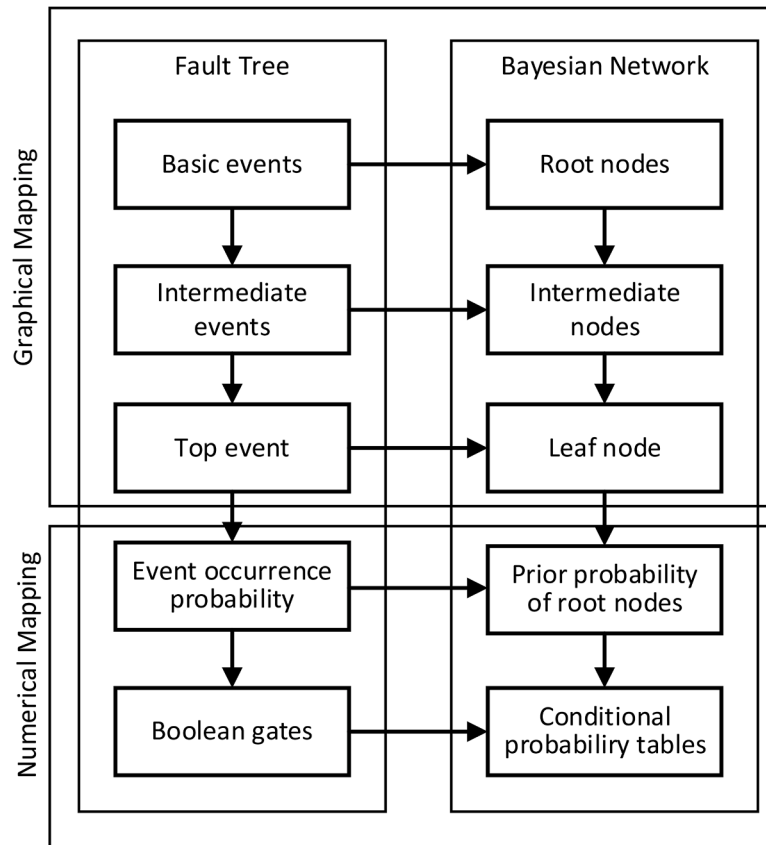


Fig. 3. Fault tree to Bayesian network mapping procedure [47].

rate  $\lambda$  caused by disasters is calculated using the previously mentioned Eqs. (1)–3. The recovery rate  $\mu$  is derived from the maintenance time of components, which is determined based on historical data and expert knowledge. Tables 2 and 3 define the temporal conditional probability

tables (CPTs) for each time-dependent component node in the DBN, describing the state transition from time  $T$  to  $T + \Delta t$  under a two-state Markov assumption (Normal operation and Failure). These temporal CPTs enable stepwise propagation of component degradation and repair

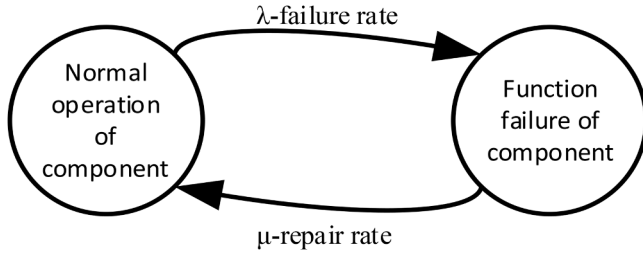


Fig. 4. Operation state transition of components influenced by disasters and recovery measures.

Table 2

Time-dependent conditional probabilities of critical components without recovery at  $T+\Delta t$ .

| Component conditions at T without recovery | Components conditions at $T+\Delta t$ |                                     |
|--------------------------------------------|---------------------------------------|-------------------------------------|
|                                            | Normal operation                      | Failure                             |
| Normal operation                           | $e^{-(\lambda \cdot \Delta t)}$       | $1 - e^{-(\lambda \cdot \Delta t)}$ |
| Failure                                    | 0                                     | 1                                   |

Table 3

Time-dependent conditional probabilities of critical components with recovery at  $T+\Delta t$ .

| Component conditions at T with recovery | Components conditions at $T+\Delta t$ |                                     |
|-----------------------------------------|---------------------------------------|-------------------------------------|
|                                         | Normal operation                      | Failure                             |
| Normal operation                        | $e^{-(\lambda \cdot \Delta t)}$       | $1 - e^{-(\lambda \cdot \Delta t)}$ |
| Failure                                 | $1 - e^{-(\mu \cdot \Delta t)}$       | $e^{-(\mu \cdot \Delta t)}$         |

over time, which is essential for generating time-varying resilience performance trajectories.

In Table 2, a component that is functional at time  $T$  fails within  $[T, T + \Delta t]$  with probability  $1 - e^{-\lambda \Delta t}$ , while a failed component remains failed, i.e.,  $P(\text{Normal}|\text{Failure}) = 0$  and  $P(\text{Failure}|\text{Failure}) = 1$ .

In Table 3, a failed component can return to normal operation within  $[T, T + \Delta t]$  according to the repair rate  $\mu$ , with  $P(\text{Normal}|\text{Failure}) = 1 - e^{-\mu \Delta t}$  and  $P(\text{Failure}|\text{Failure}) = e^{-\mu \Delta t}$ .

By iterating these CPTs across time steps and propagating component states through correlations links (noisy-OR/AND), the DBN produces time-iterative probabilities of system-level service loss, from which resilience performance trajectories and recovery-strategy comparisons are computed.

### 3.4. Model validation

The validation of DBNs is carried out using sensitivity analysis, which assesses the uncertainty in the model's output and attributes it to variations in the input variables. If the methodology is sound and the reasoning is logical, the sensitivity analysis must reflect at least one of the following three axioms [48]:

- Increasing or decreasing the prior probability of each parent node, resulting in a corresponding increase or decrease in the posterior probability of the associated child nodes.
- The magnitude of the effect of evidence  $x$  on changing the probability of the system is always greater than the effect of any subset of  $x$  (i.e.,  $x - yx - yx - y$ , where  $y \subseteq x$  and  $y \subseteq x$ ).
- The magnitude of the effect of evidence  $x$  on a change in the probability of the system is always greater than the effect of  $x - y$  ( $y \in x$ ) on it.

The proposed model is solved through probabilistic inference on an

integrated FT-DBN model. First, the FT is analysed using an analytical approach to compute the occurrence probabilities of basic events, minimal cut sets, and top events, which are then used to parameterise the prior probabilities and baseline conditional probability tables (CPTs) of the DBN. Next, the DBN is evaluated via time-slice propagation under the Markov assumption, using degradation and recovery transition CPTs to update node states over time and generate resilience performance trajectories under different rainfall intensities and recovery strategies. The fault tree (FT) is evaluated in Python using standard scientific libraries (NumPy/SciPy/Pandas). The DBN construction and inference were implemented in GeNIe 5.0 (BayesFusion), which performs Bayesian-network belief updating to obtain posterior probabilities for component and system states.

## 4. Case study: model development for resilience performance

The proposed resilience performance model is demonstrated based on a case study of the Water-Electricity-Road (WER) system in a typical community located in China. Based on the district's characteristics, common disasters include heavy rain, thunderstorms, and extreme hot and cold temperatures. In this case study, the WER system ensures electricity supply, water distribution, and road access to four buildings. Among them, buildings 1, 2, and 3 are high-rise structures (Fig. 5).

Given the large number of real assets, this case study abstracts the key functional components that govern service delivery and cross-sector interdependency. The electricity distribution system is represented by the main underground substation, above-ground substation, and underground high-voltage and low-voltage cables, and follows a radial configuration in which power is supplied from the municipal grid to the main underground substation, transmitted via 10 kV cables to the above-ground substation, and then stepped down to 3 kV for distribution to the buildings. The water distribution system comprises main pipelines, sub-pipelines, valve control stations, and pumps, and follows a looped configuration where pressurised municipal water is conveyed through the main network to branch pipelines supplying the buildings. Buildings 1–3 are equipped with pumps to maintain adequate pressure for upper floors. The road network is simplified as 2 main road segments with intersections and traffic lights. Interdependency is modelled primarily as physical interdependency, where electricity enables pump operation and thus water service, while road accessibility affects the ability to access and restore disrupted components during rainfall scenarios.

### 4.1. Fault Tree (FT) model

FT analysis is used to identify potential causes of failure under non-disaster scenarios. In this study, 3 top events are considered: water supply failure, electricity supply failure, and road network failure. These top events represent the logical outcomes of system failures and are illustrated in Fig. 6. The failure probabilities of the BEs contributing to the FT are summarised in Table 4. Some BEs have fixed failure probabilities (e.g., B1, B2), while others have time-dependent failure probabilities calculated using the Eq. (4), (e.g., B12, B13). The corresponding intermediate events (IEs) in the FT are presented in Table 5.

### 4.2. Corresponding Dynamic Bayesian Network model

The FT illustrates the failure propagation path within the WER system and identifies the BEs under non-disaster scenarios. However, under disaster conditions, the probability of BEs occurrence significantly increases. For instance, in a rainfall scenario, Fig. 7 mechanical failure of the pump in building 1 (BE5) can occur due to high humidity levels and waterlogging-induced submersion. Moreover, the probability of such failures tends to increase over time.

According to the conversion algorithm from FT to DBN described in Section 3.3, the corresponding DBN is shown in Fig. 8. The basic events that can be triggered by rainfall including B5, B6, B7, B9, B17, B18, and

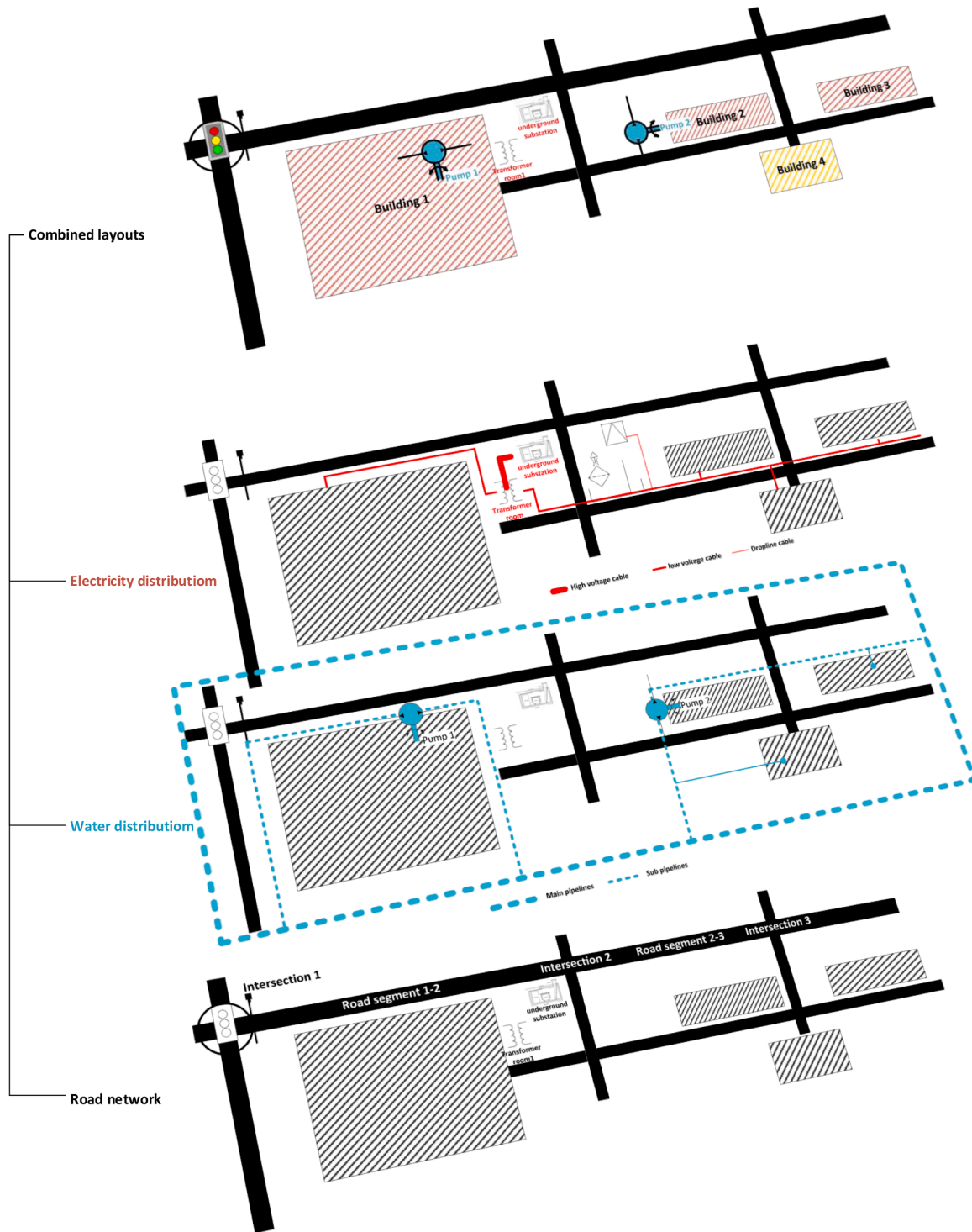


Fig. 5. The layout of WER systems in the case study.

B19 are transferred to time dependent nodes of DBN (Table 9). In contrast, BEs unaffected by rainfall are represented as constant nodes.

For the time dependent nodes, the failure rate ( $\lambda$ ) induced by rainfall is calculated using Eq. (1) or Eq. (2). Specifically, for equipment-related failures, factors such as relative humidity (RH) and water submersion levels (D) are used to determine failure probability, as defined in Eq. (1). For road segments, rainfall intensity (RI) is used to calculate failure probability according to Eq. (2). This study evaluates the resilience performance of the WER system under moderate rainfall (MR), heavy rainfall (HR), and extreme heavy rainfall (EHR) scenarios. The

parameters for three rainfall scenarios are provided in Table 6. Tables 7 and 8 provide two illustrative examples of failure rate ( $\lambda$ ) calculations under the three rain scenarios for BE7 and BE18. Finally, the failure rates ( $\lambda$ ) of all time dependent BEs affected by rainfall and the recovery rates ( $\mu$ ) are summarised in Table 9. The recovery rates ( $\mu$ ) are estimated by the community manager in the case study.

The DBN is used to simulate the time-evolving normal operation probabilities of the WER system under rainfall scenarios. The operation probabilities of time-dependent nodes are calculated using the equations provided in Tables 2 and 3, considering both repair and no-repair

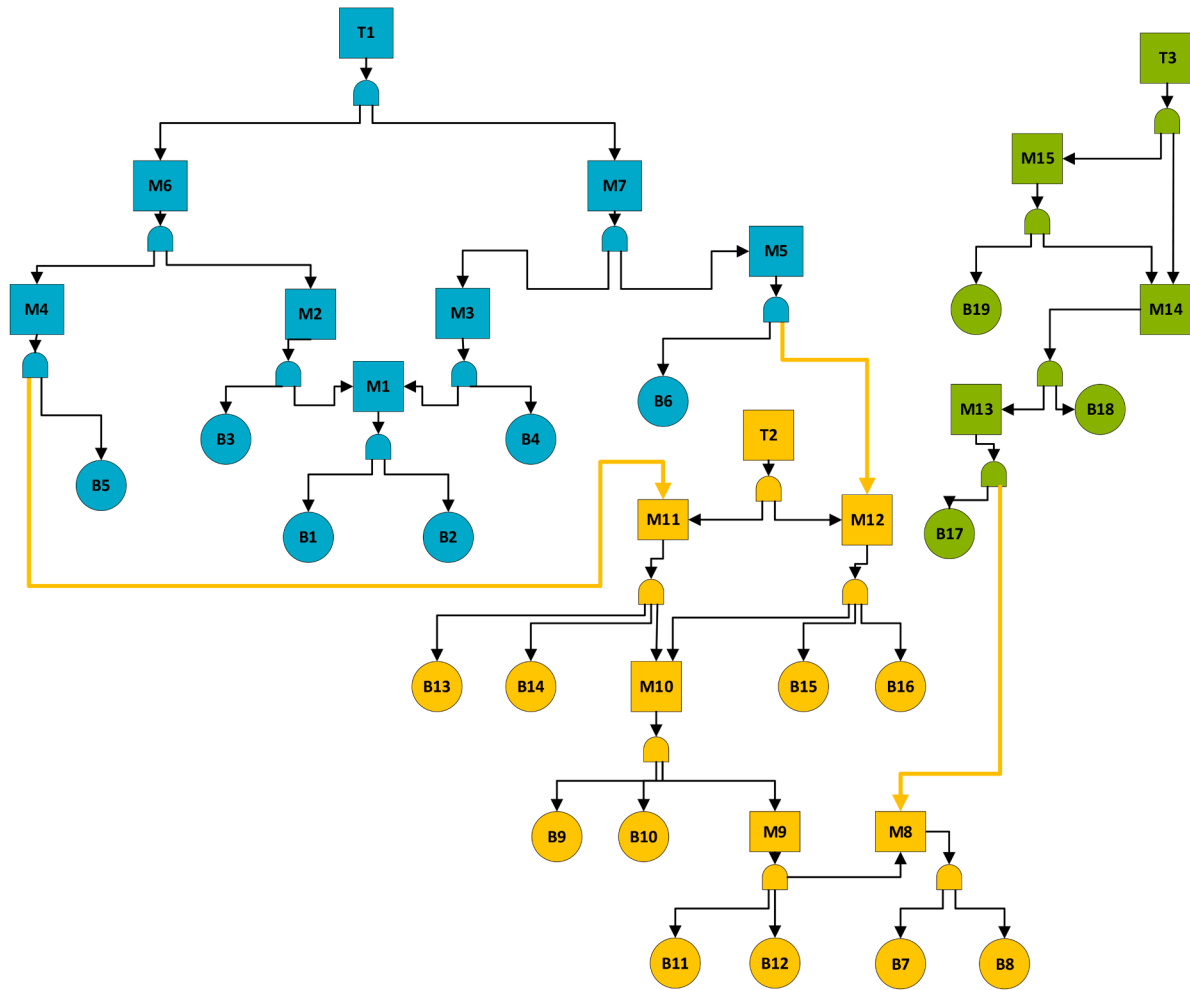


Fig. 6. FT of WER systems in the non-disaster scenario.

scenarios. For constant nodes, the failure probabilities obtained from the FT analysis are used. The conditional probabilities of each leaf nodes in the DBN are derived based on the behaviour of the logical gates, as shown in Eq. (7) and 8.

Validation is a critical step in demonstrating that the model provides a reasonable representation of the actual system. To support this, a sensitivity analysis was conducted as a means of partial validation. At a minimum, the model is expected to satisfy at least one of the three axioms outlined in Section 3.4.

## 5. Results

### 5.1. Operation failure results in non-disaster scenarios using FT analysis

There are 19 basic events in the Fault Tree model of the case study. As shown in Table 10, most of the probabilities are  $<0.1\%$ , which means that the failure probability of single events is small. The highest probability is related to B3 and B4 which is equal to  $0.26\%$ , representing corrosion or mechanical damage to sub-pipelines for buildings 1 and building 2–4. The intermediate nodes (M6 & M7), and (M2 & M3) have probabilities significantly higher than other intermediate events,  $0.95\%$  and  $0.48\%$ , respectively—representing loss of water supply for Buildings 1–4 and loss of water supply from sub-pipelines 1 and 2. Among the top events, T1, T2, and T3, the probability of water supply failure (T1) is significantly higher at  $1.89\%$ , compared to electricity supply failure ( $0.67\%$ ) and limited road network access ( $0.58\%$ ).

### 5.2. Resilience performance of water, electricity, and road systems under disaster scenarios using DBN analysis

Assuming that the WER systems are subjected to moderate, heavy, and extreme rain scenarios, the parameters used for the DBN analysis are provided in Table 6–9. The resilience performance of the WER system under three different rain scenarios, both with and without repairs, is shown in Fig. 8 and Fig. 9, respectively. In these figures, the vertical axis represents the probability of normal operation for nodes T1 (water supply), T2 (electricity supply), and T3 (road accessibility) under different rainfall scenarios. The horizontal axis shows the time steps in hour after the disaster (Figs. 8–9). For instance, time step 24 represents the situation 24 h after the occurrence of the disaster.

As shown in Fig. 8, across all rainfall scenarios, the road system consistently exhibits the most rapid and severe degradation, especially under extreme heavy rain (EHR) scenario. By time step 24, the normal operation probability of the road system under EHR reaches  $0.01\%$ , dropping from  $99.42\%$  at the start. The normal operation probability of electricity system under EHR falls to  $42.80\%$  at time step 24, while the water system maintains slightly lower probability at  $35.08\%$  under the same scenario. Under moderate rain (MR) scenario, all three systems show a relatively slower decline in performance. The final values of normal operation probability at time step 24 for road, water, and electricity networks are  $18.88\%$ ,  $68.54\%$ , and  $74.13\%$ , respectively. The steeper slope in the road system highlights its heightened sensitivity even under moderate conditions. The results indicate that the road system is the most fragile, particularly under extreme rainfall.

**Table 4**

Basic events (BEs) of the FT and their failure probability.

| Infrastructure systems          | Components                      | Identifiers of basic events | Basic events                                                                          | Shape ( $\beta$ ) | Scale ( $\eta$ ) | Failure probability |
|---------------------------------|---------------------------------|-----------------------------|---------------------------------------------------------------------------------------|-------------------|------------------|---------------------|
| Water distribution system       | Main pipelines                  | B1                          | Corrosion or mechanical damage to main pipes                                          | 1.5               | 45               | 0.09 %              |
|                                 | Main valves control station     | B2                          | Corrosion or mechanical damage to valves                                              | 1.3               | 35               | 0.13 %              |
|                                 | Sub pipelines                   | B3                          | Corrosion or mechanical damage to sub pipelines for building 1                        | 1.1               | 10               | 0.26 %              |
|                                 |                                 | B4                          | Corrosion or mechanical damage to sub pipelines for buildings 2–4                     | 1.1               | 10               | 0.26 %              |
|                                 | Water pumps                     | B5                          | Mechanical damage to the pump for building 1                                          | 1.15              | 15               | 0.14 %              |
|                                 |                                 | B6                          | Mechanical damage to the pump for buildings 2–4                                       | 1.15              | 15               | 0.14 %              |
| Electricity distribution system | Main underground substation     | B7                          | Mechanical damage to critical equipment                                               | 1.2               | 20               | 0.07 %              |
|                                 |                                 | B8                          | Thermal failure of critical equipment                                                 | \                 | \                | 0.09 %              |
|                                 | Above-ground substation         | B9                          | Mechanical damage to critical equipment in Substation                                 | 1.2               | 25               | 0.06 %              |
|                                 |                                 | B10                         | Thermal failure of critical equipment in Substation                                   | \                 | \                | 0.09 %              |
|                                 | Underground high-voltage cables | B11                         | Corrosion or mechanical damage to underground high voltage cables to transformer room | 1.2               | 30               | 0.04 %              |
|                                 |                                 | B12                         | Thermal failure of underground high voltage cables to transformer room                | \                 | \                | 0.04 %              |
|                                 | Underground low-voltage cables  | B13                         | Corrosion or mechanical damage to underground low voltage cables for building 1       | 1.2               | 25               | 0.05 %              |
|                                 |                                 | B14                         | Thermal failure of underground low voltage cables for building 1                      | \                 | \                | 0.05 %              |
|                                 |                                 | B15                         | Corrosion or mechanical damage to underground low voltage cables for buildings 2–3    | 1.2               | 25               | 0.05 %              |
|                                 |                                 | B16                         | Thermal failure of underground low voltage cables for buildings 2–3                   | \                 | \                | 0.05 %              |
| Road network                    | Traffic lights Intersections    | B17                         | Mechanical damage to the traffic lights                                               | 1.1               | 15               | 0.13 %              |
|                                 |                                 | B18                         | Low accessibility of road segment 1                                                   | \                 | \                | \                   |
|                                 |                                 | B19                         | Low accessibility of road segment 2                                                   | \                 | \                | \                   |

**Table 5**

List of intermediate events in the FT.

| Infrastructure systems          | ID of intermediate events | Intermediate events                                                        |
|---------------------------------|---------------------------|----------------------------------------------------------------------------|
| Water distribution system       | M1                        | Loss of water supply from the main pipes                                   |
|                                 | M2                        | Loss of water supply from sub-pipelines 1                                  |
|                                 | M3                        | Loss of water supply from sub-pipelines 2                                  |
|                                 | M4                        | Failure of pump 1                                                          |
|                                 | M5                        | Failure of pump 2                                                          |
|                                 | M6                        | Loss of water supply for building 1                                        |
|                                 | M7                        | Loss of water supply for buildings 2–4                                     |
|                                 | M8                        | Power outage at the main underground substation                            |
| Electricity distribution system | M9                        | Power outage at the high-voltage cables                                    |
|                                 | M10                       | Power outage at the above-ground substation                                |
|                                 | M11                       | Power outage at the low-voltage cables 1 (Power outage at buildings 1)     |
|                                 | M12                       | Power outage at the low-voltage cables 2–4 (Power outage at buildings 2–4) |
| Road network                    | M13                       | Traffic light malfunction                                                  |
|                                 | M14                       | Low accessibility to the road segment 1                                    |
|                                 | M15                       | Low accessibility to road segment 2                                        |

Therefore, prioritising road recovery in mitigation strategies is important and is likely to yield cascading benefits for the dependent electricity and water networks.

Assuming the WER systems are impacted by rainfall, and the recovery measures for WER system is implemented 6 hrs after rainfall occurs (Recovery Strategy 1), the corresponding recovery rates are provided in Table 9. The simulated performance of WER system after implementing the recovery is shown in Fig. 9. The lowest performance value (normal operation probability) of the water distribution system under MR, HR, and EHR is 90.52 %, 84.63 %, and 78.26 %, respectively.

As shown in Fig. 9, the lowest normal operation probability of the electricity system under MR, HR, and EHR is 92.97 %, 87.97 %, and 82.51 %, respectively. The lowest performance values of road networks under MR, HR, and EHR is 69.29 %, 37.04 %, and 12.47 %, respectively. It is observed that the rainfall disaster scenario has the largest effect on the road network, particularly under the heavy rain condition. However, the road network demonstrates a faster recovery compared to the water and electricity distribution networks, outperforming both by time step 11. Between the water distribution and electricity networks, the performance of water distribution system is more impacted by the rainfall, as it relies on electricity to operate. In other words, the water distribution system is impacted both directly by rainfall and indirectly through failures in the electricity network.

### 5.3. Impact of two alternative recovery strategies on resilience performance of water, electricity, and road systems

Different recovery strategies could be employed to restore the systems under moderate rain (MR), heavy rain (HR), and extreme heavy rain (EHR) scenarios. In this research, two recovery strategies are employed. The first strategy (S1) is a general recovery approach, in which all systems are repaired 6 h after the onset of rainfall. The decision to delay recovery actions by 6 h is based on the characteristics of the case study area. According to data from the Beijing Weather Station, the average duration of a single short rainfall event ranges from 1 to 4 h, and rarely exceeds 6 h. This suggests that recovery efforts can typically begin after 6 h without interference from ongoing rainfall. The second strategy (S2) involves a sequential recovery: the road network, electricity distribution network, and water distribution system are restored at time steps 4, 5, and 6, respectively. This sequence is based on earlier analysis indicating that the road network is easy to recover, and that the electricity network is essential for the operation of the water distribution system.

The simulated values of the normal operation probability of WER system (Node T1, T2, and T3) under two recovery strategies S1 and S2 is shown in Figs. 10–12. The 3D line plots clearly illustrate these trends, revealing early-stage differences between S1 and S2 under MR, HR and

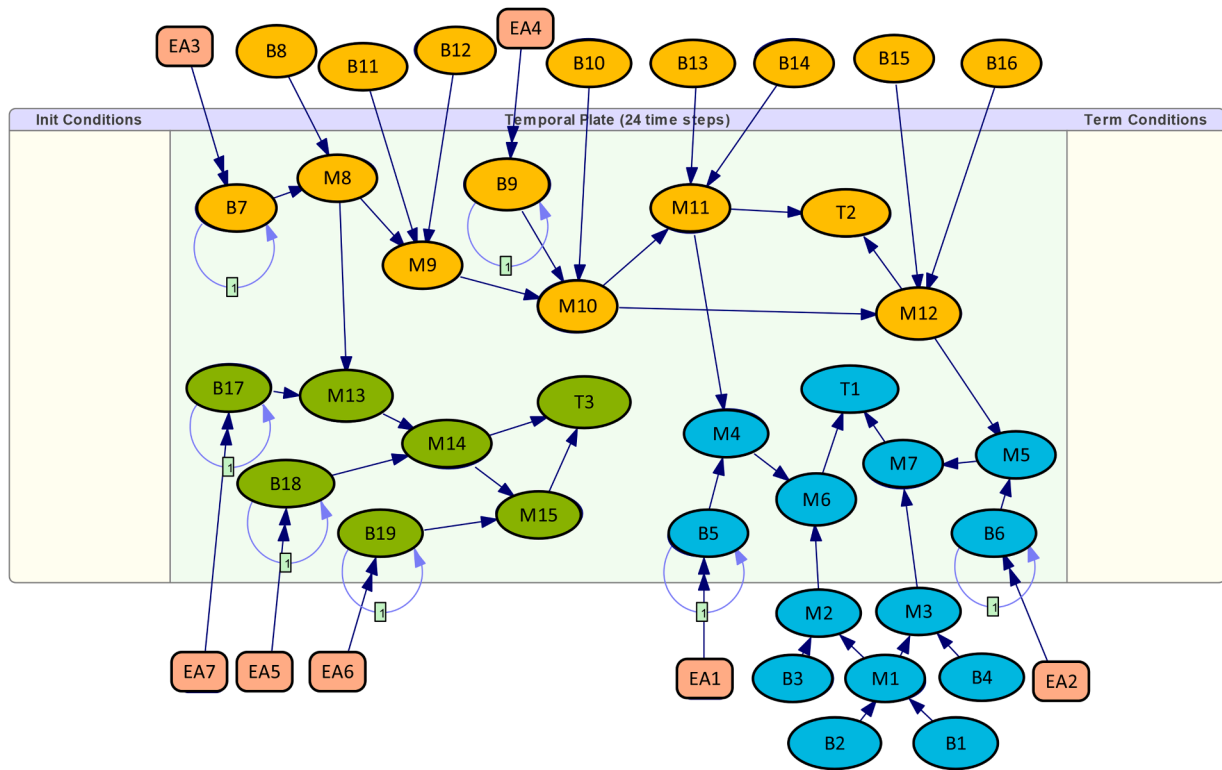


Fig. 7. DBN model of WER systems under rainfall scenario.

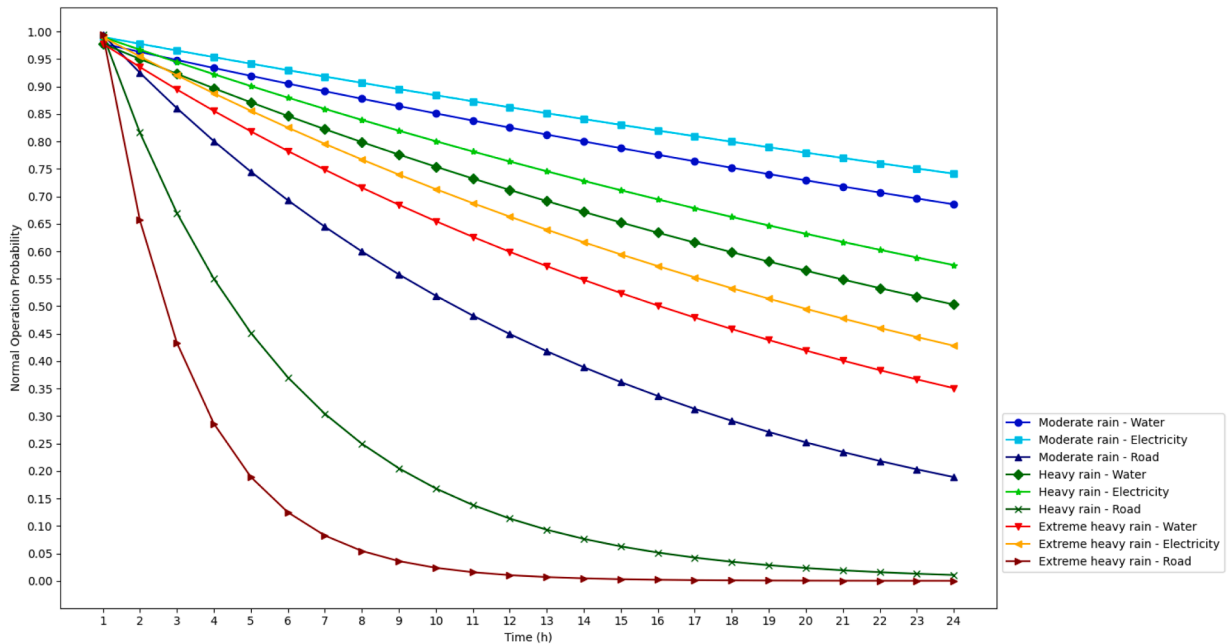


Fig. 8. The normal operation probability of the WER system under three rain scenarios without recovery measures.

EHR conditions, and highlighting the benefits of staggered, prioritised recovery strategies in severe weather scenarios.

Fig. 10 shows that the recovery Strategy 1 results in lower performance compared to Strategy 2 for the water distribution network. Fig. 10 shows that the normal operation probability of the water distribution network steadily declines from time step 1 to time step 6 (S1) and to time step 4 (S2), followed by a partial recovery that extends to time step 24. For instance, under the EHR scenario, the normal operation probability for water supply drops sharply from 97.80 % at time step 1 to

approximately 78.26 % (S1) and 81.69 % (S2) by time step 6, before stabilising around 89.28 % (S1) and 91.01 % (S2) at the time step 24. Sequenced recovery Strategy 2 demonstrates improved performance recovery, particularly under the HR and EHR scenarios. For example, in the HR scenario, Strategy 2 achieves a final normal operation probability of 93.44 %, compared to 92.27 % under Strategy 1. A similar trend is observed in the EHR scenario, the normal operation probability for water supply reaches 91.01 % (S2) and 89.28 % (S1). In MR scenario, performance remains relatively stable due to minimal initial disruption,

**Table 6**

The parameters of the three rainfall scenarios: moderate rainfall (MR), heavy rainfall (HR), and extreme heavy rainfall (EHR).

| Disaster scenarios | Influencing factors |         |             |         |                |         |
|--------------------|---------------------|---------|-------------|---------|----------------|---------|
|                    | D_real (m)          | D_norm  | RH_real (%) | RH_norm | RI_real (mm/h) | RI_norm |
| Moderate rain      | 0.3                 | 0.18182 | 0.94        | 0.44444 | 25             | 0.2     |
| Heavy rain         | 0.4                 | 0.27273 | 0.96        | 0.66667 | 40             | 0.5     |
| Extreme heavy rain | 0.5                 | 0.36363 | 0.975       | 0.83333 | 60             | 0.9     |

**Table 7**

Failure rate  $\lambda$  for B7 and parameters of rainfall used for B7.

| Disaster scenarios | Influencing factors |         |             |         | Failure rate $\lambda$ |
|--------------------|---------------------|---------|-------------|---------|------------------------|
|                    | D_real (m)          | D_norm  | RH_real (%) | RH_norm |                        |
| Moderate rain      | 0.3                 | 0.18182 | 0.94        | 0.44444 | 0.00287                |
| Heavy rain         | 0.4                 | 0.27273 | 0.96        | 0.66667 | 0.00541                |
| Extreme heavy rain | 0.5                 | 0.36364 | 0.975       | 0.83333 | 0.00835                |

**Table 8**

Failure rate  $\lambda$  for B18 and parameters of rainfall used for B18.

| Disaster scenarios | Influencing factors |         | Failure rate $\lambda$ |
|--------------------|---------------------|---------|------------------------|
|                    | RI_real(mm/h)       | RI_norm |                        |
| Moderate rain      | 25                  | 0.27273 | 0.02136                |
| Heavy rain         | 40                  | 0.54545 | 0.06043                |
| Extreme heavy rain | 60                  | 0.90909 | 0.13002                |

with both strategies achieving similar final values of approximately 94.94 % (S1) and 95.59 % (S2) at time step 24.

Fig. 11 shows that the recovery Strategy 1 results in lower performance compared to Strategy2 for the electricity distribution network, particularly under HR and EHR conditions. For instance, under the EHR scenario, the normal operation probability under Strategy 1 decreases from 99.01 % at time step 1 to a 82.51 % at time step 6, before gradually recovering to 91.32 % by time step 24. In contrast, sequenced recovery Strategy 2 consistently achieves higher normal operation probabilities across both HR and EHR scenarios. Under the EHR scenario, the electricity supply reaches normal operation probability of 93.09 % under Strategy 2 by time step 24, slightly outperforming Strategy 1. In the HR scenario, the final normal operation probabilities are 95.16 % (S2) and 93.98 % (S1), with Strategy 2 demonstrating a noticeably faster recovery during the mid-phase.

Fig. 12 shows that the recovery Strategy 1 results in lower

performance compared to Strategy2 for the road networks. Among the three infrastructure sectors, the road network (Fig. 12) exhibits the most rapid performance decline and recovery, particularly under the EHR scenario. Under Strategy 1, where recovery begins uniformly at time step 6, the normal operation probability of the road network drops sharply from 99.42 % at time step 1 to 12.47 % at time step 6, indicating severe degradation. However, performance rebounds quickly, recovering to 94.95 % by time step 24. Strategy 2 leads to improved mid-phase performance across all rainfall levels. Under the EHR scenario, the normal operation probability at time step 6 is significantly higher at 68.24 %, compared to 12.47 % under Strategy 1. By the final time step 24, both S1 and S2 converge to similar outcomes (e.g., 96.30 % under S2 versus 94.95 % under S1), though Strategy 2 maintains consistently higher performance throughout the recovery period. Under MR and HR conditions, the differences between the two strategies are subtler but still evident. Strategy 2 yields slightly higher normal operation probabilities at nearly every time step. Under the HR scenario, Strategy 2 reaches normal operation probability of 97.48 % by time step 24, compared to 96.58 % under Strategy 1.

#### 5.4. Performance of different components of water, electricity, and road systems under disaster scenarios and recovery strategies

Figs. 13–15 present the failure probabilities of critical components — similar to occurrence probability of BEs — within the water, electricity, and road networks under Moderate Rain (MR), Heavy Rain (HR), and Extreme Heavy Rain (EHR) scenarios, based on general Strategy 1.

**Table 10**

Failure probability based on the Fault Tree.

| ID of Basic events | Failure probability | ID of Intermediate events | Failure probability | ID of Top events | Failure probability |
|--------------------|---------------------|---------------------------|---------------------|------------------|---------------------|
| B1                 | 0.09 %              | M1                        | 0.22 %              | T1               | 1.89 %              |
| B2                 | 0.13 %              | M2                        | 0.48 %              | T2               | 0.67 %              |
| B3                 | 0.26 %              | M3                        | 0.48 %              | T3               | 0.58 %              |
| B4                 | 0.26 %              | M4                        | 0.47 %              |                  |                     |
| B5                 | 0.14 %              | M5                        | 0.47 %              |                  |                     |
| B6                 | 0.14 %              | M6                        | 0.95 %              |                  |                     |
| B7                 | 0.07 %              | M7                        | 0.95 %              |                  |                     |
| B8                 | 0.09 %              | M8                        | 0.16 %              |                  |                     |
| B9                 | 0.06 %              | M9                        | 0.08 %              |                  |                     |
| B10                | 0.09 %              | M10                       | 0.24 %              |                  |                     |
| B11                | 0.04 %              | M11                       | 0.34 %              |                  |                     |
| B12                | 0.04 %              | M12                       | 0.34 %              |                  |                     |
| B13                | 0.05 %              | M13                       | 0.29 %              |                  |                     |
| B14                | 0.05 %              | M14                       | 0.29 %              |                  |                     |
| B15                | 0.05 %              | M15                       | 0.29 %              |                  |                     |
| B16                | 0.05 %              |                           |                     |                  |                     |
| B17                | 0.13 %              |                           |                     |                  |                     |
| B18                | \                   |                           |                     |                  |                     |
| B19                | \                   |                           |                     |                  |                     |

**Table 9**

Failure rate ( $\lambda$ ) and recovery rate ( $\mu$ ) for the time-dependent nodes under moderate rainfall (MR), heavy rainfall (HR), and extreme heavy rainfall (EHR).

| Disaster scenarios | Disaster may trigger basic event                                         | Trigger reason                                                                             | Failure rate $\lambda$ |                |                 | Recovery rate $\mu$ |
|--------------------|--------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|------------------------|----------------|-----------------|---------------------|
|                    |                                                                          |                                                                                            | $\lambda_{MR}$         | $\lambda_{HR}$ | $\lambda_{EHR}$ |                     |
| Rainfall scenarios | B5: Mechanical damage of the pump for building 1                         | Water logging submerge pump                                                                | 0.00132                | 0.00242        | 0.00374         | 0.06667             |
|                    | B6: Mechanical damage of the pump for building 2&3                       | Water logging submerge pump                                                                | 0.00155                | 0.00285        | 0.00439         | 0.06667             |
|                    | B7: Corrosion or mechanical damage of critical equipment                 | Water logging submerges the critical equipment; Humidity induced short-circuited equipment | 0.00287                | 0.00541        | 0.00835         | 0.04348             |
|                    | B9: Corrosion or mechanical damage of critical equipment in substation 1 | Water logging submerges the critical equipment; Humidity induced short-circuited equipment | 0.00342                | 0.00641        | 0.00988         | 0.04348             |
|                    | B17: Mechanical damage of the traffic light                              | Rainfall damage to lights (LED or Incandescent                                             | 0.00119                | 0.00267        | 0.00417         | 0.0738              |
|                    | B18: low access of intersection 1                                        | Intense rainfall and waterlogging in the road segment 1                                    | 0.02136                | 0.06043        | 0.13002         | 0.625               |
|                    | B19: low access of interSection 2                                        | Intense rainfall and waterlogging in the road segment 2                                    | 0.02136                | 0.06043        | 0.13002         | 0.625               |
|                    |                                                                          |                                                                                            |                        |                |                 |                     |
|                    |                                                                          |                                                                                            |                        |                |                 |                     |

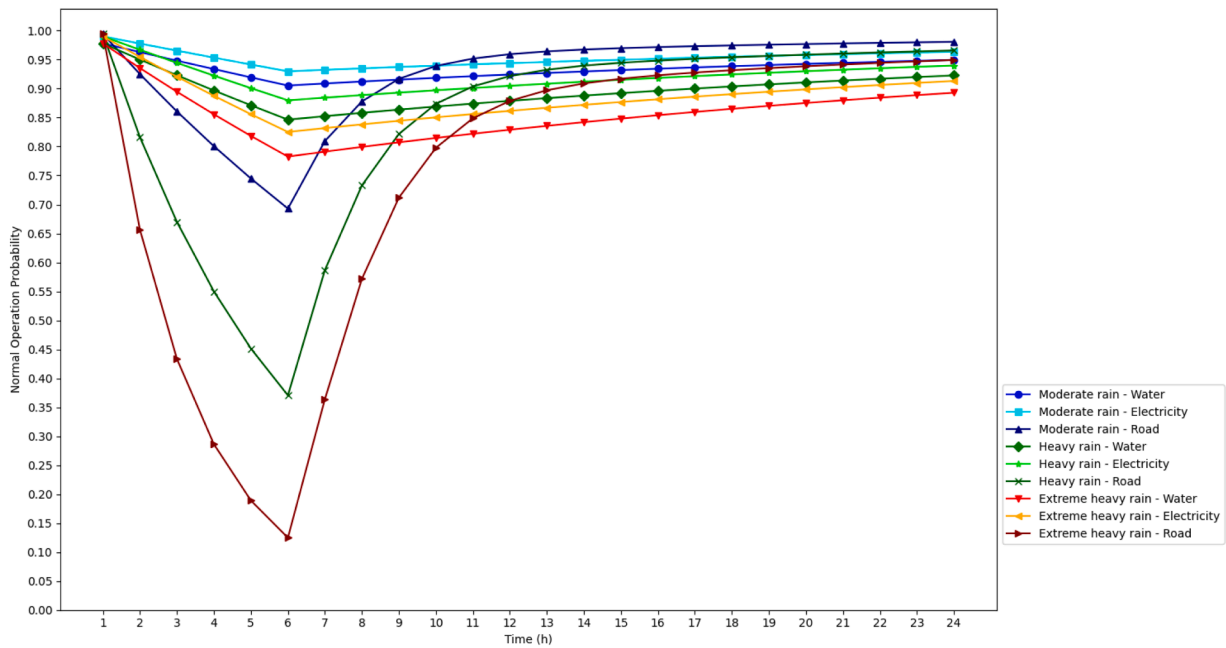


Fig. 9. The normal operation probability of the WER system under three rain scenarios with recovery strategy 1.

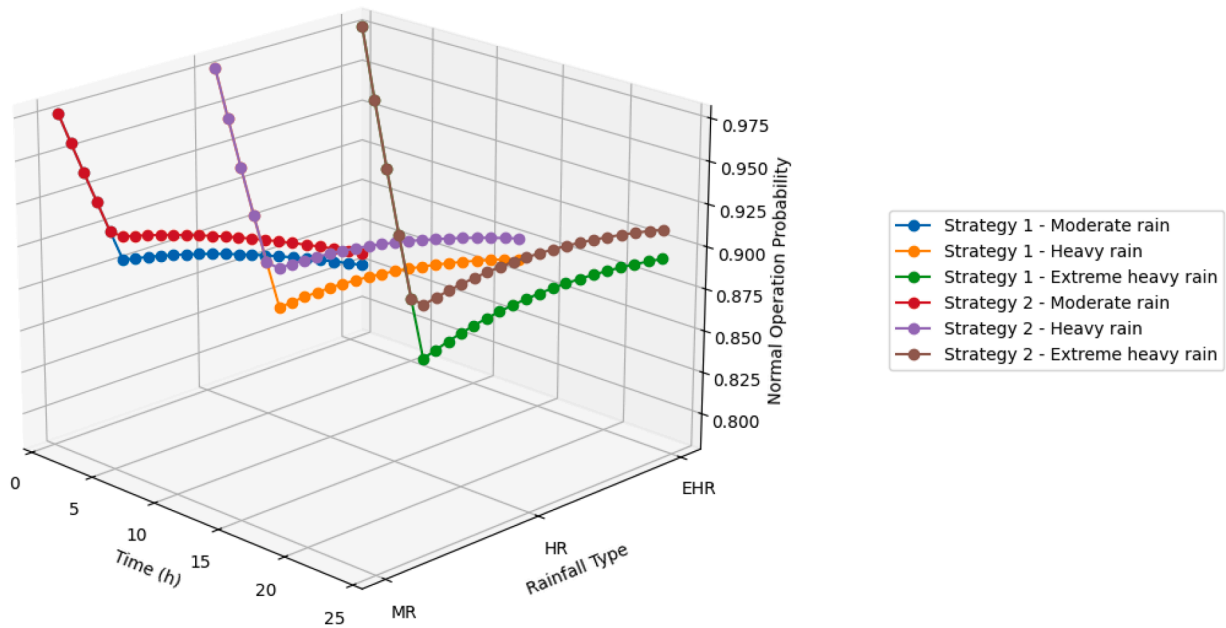


Fig. 10. The normal operation probability of the water distribution system (node T1) under three different rain scenarios and two recovery strategies.

For the water distribution system (Fig. 13), BE5 exhibits higher occurrence probabilities than BE6 across all rainfall scenarios. Prior to the implementation of recovery measures at time step 6, the occurrence probabilities of both BE5 and BE6 increase rapidly. After this point, the growth in occurrence probability slows, with the highest values observed under the HER scenario. By the time step 24, the occurrence probability of BE6 under the EHR scenario reaches 0.69 %, while BE5 reaches 0.59 %. Under the HR scenario, BE5 and BE6 reach 0.40 % and 0.46 %, respectively. In the MR scenario, the values are lower, with BE5 at 0.23 % and BE6 at 0.27 %.

The occurrence probabilities of electricity-related basic events BE7 and BE9 (Fig. 14) exhibit clear sensitivity to rainfall intensity over time. Initially, all three scenarios start with low probabilities, BE7 is 0.07 % and BE9 is 0.06 %. By time step 6, notable differences emerge. Under MR

scenarios, BE7 and BE9 increase to 1.49 % and 1.75 %, respectively; under HR scenarios, they rise to 2.73 % for BE7 and 3.21 % for BE9. In the EHR scenario, the probabilities peak at 4.15 % for BE7 and 4.88 % for BE9, highlighting the compounding stress induced by severe rainfall. Across all time steps, BE9 consistently maintains a higher occurrence probability than BE7, indicating greater vulnerability within the electricity network. By time step 24, partial recovery is evident: under MR, the probabilities decline to 0.68 % (BE7) and 0.80 % (BE9); under HR, to 1.25 % (BE7) and 1.47 % (BE9), respectively. Under EHR scenarios, however, high probabilities persist, with BE7 and BE9 at 1.89 % and 2.23 % respectively, underscoring the impact of extreme rainfall even after intervention measures are assumed.

The occurrence probabilities of road network-related basic events (Fig. 15) increase rapidly, peaking early in the simulation before steadily

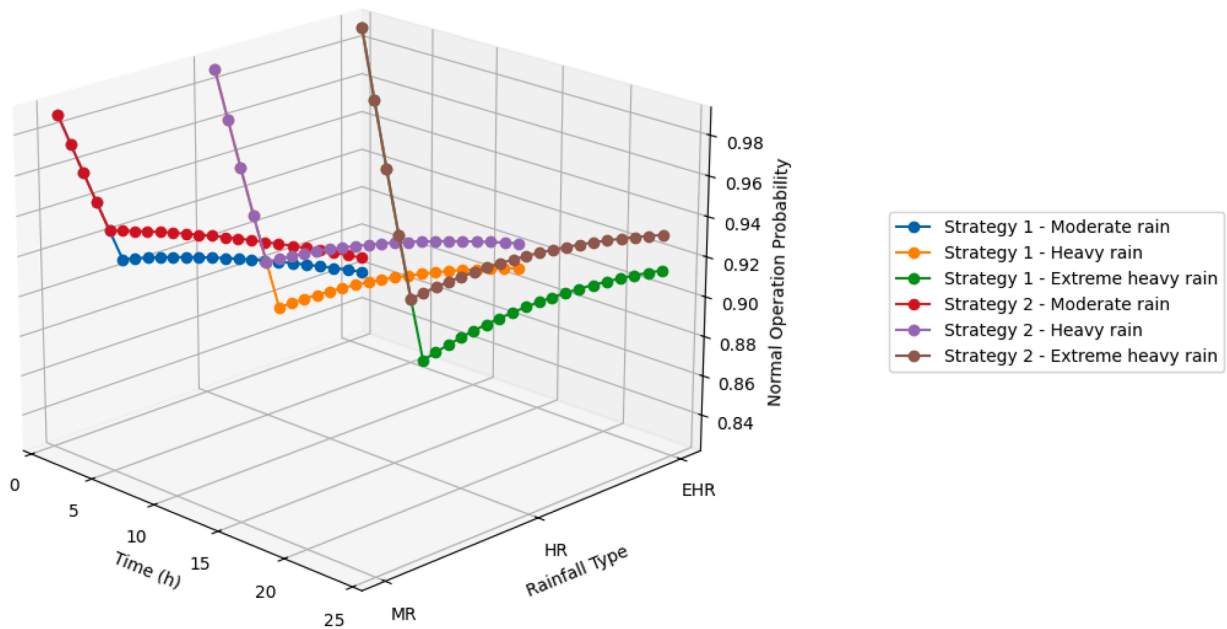


Fig. 11. The normal operation probability of the electricity distribution system (node T2) under three different rain scenarios and two recovery strategies.

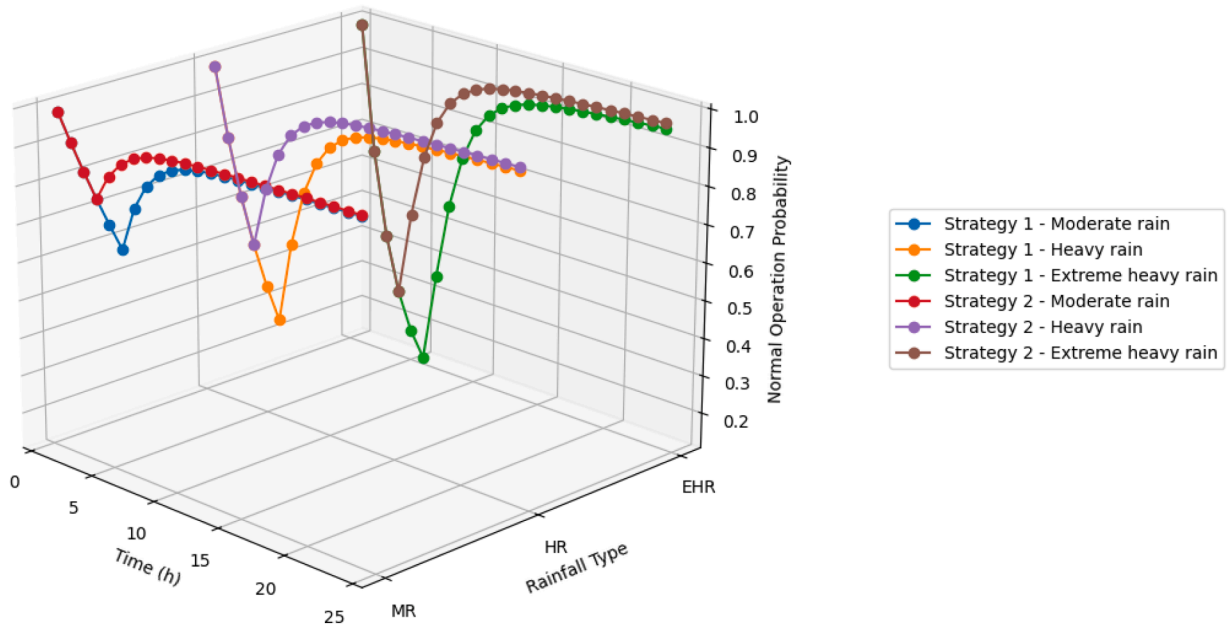


Fig. 12. The normal operation probability of the road networks (node T3) under three different rain scenarios and two recovery strategies.

declining toward the end. At time step 6, BE17, BE18, and BE19 exhibit sharp peaks across all rainfall scenarios, reflecting the moment of maximum system stress prior to the implementation of recovery interventions. Under the EHR scenario, the occurrence probabilities for BE18 and BE19 peak to 47.80 %, significantly exceeding the values observed under the HR scenario, 26.07 % for both events, and surpassing those under the MR scenario, where both reach 10.13 %. BE17, which starts from an initial value of 0.13 % across all scenarios, also reaches its respective peak at time step 6: 2.18 % for EHR, 1.45 % for HR, and 0.72 % for MR. By time step 24, after recovery efforts have been implemented, the occurrence probabilities of all three basic events decline substantially, indicating the effectiveness of the intervention strategy over time.

## 6. Discussions

Based on the operation failure results under non-disaster scenarios, the probability of corrosion or mechanical damage in the sub-pipelines supplying water to Buildings 1 and 2–4 is relatively high. This is primarily attributed to the earlier construction time of the pipelines in the case study, leading to increased ageing and fatigue of sub-pipeline components. Furthermore, under normal conditions, the probability of water supply failure (T1) is 1.89 %, which is notably higher than that of electricity supply failure (0.67 %) and road network low accessibility (0.58 %).

The resilience performance results under disaster scenarios highlight several insights in this case study. Notably, extreme heavy rainfall imposes significantly greater stress on WER systems than heavy or

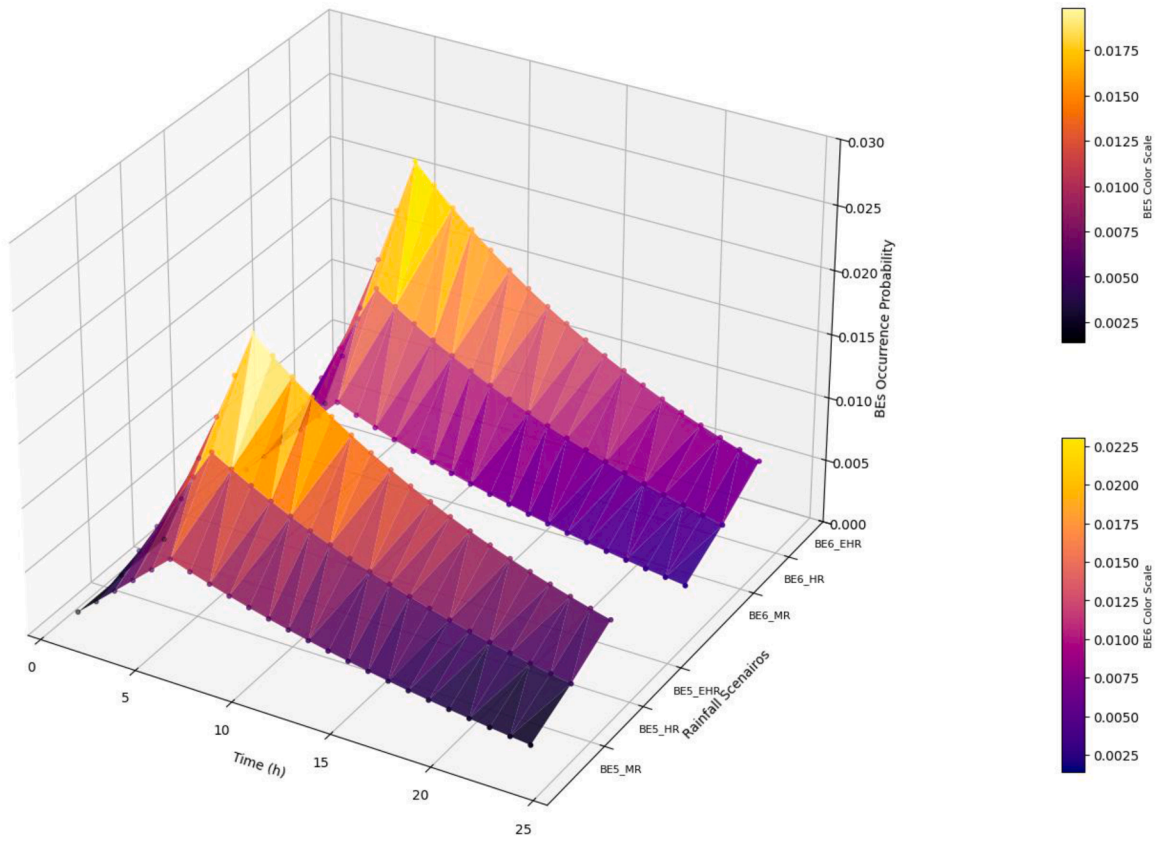


Fig. 13. The occurrence probability of BE5 and BE6 of the water distribution system under three rainfall scenarios with recovery strategy 1.

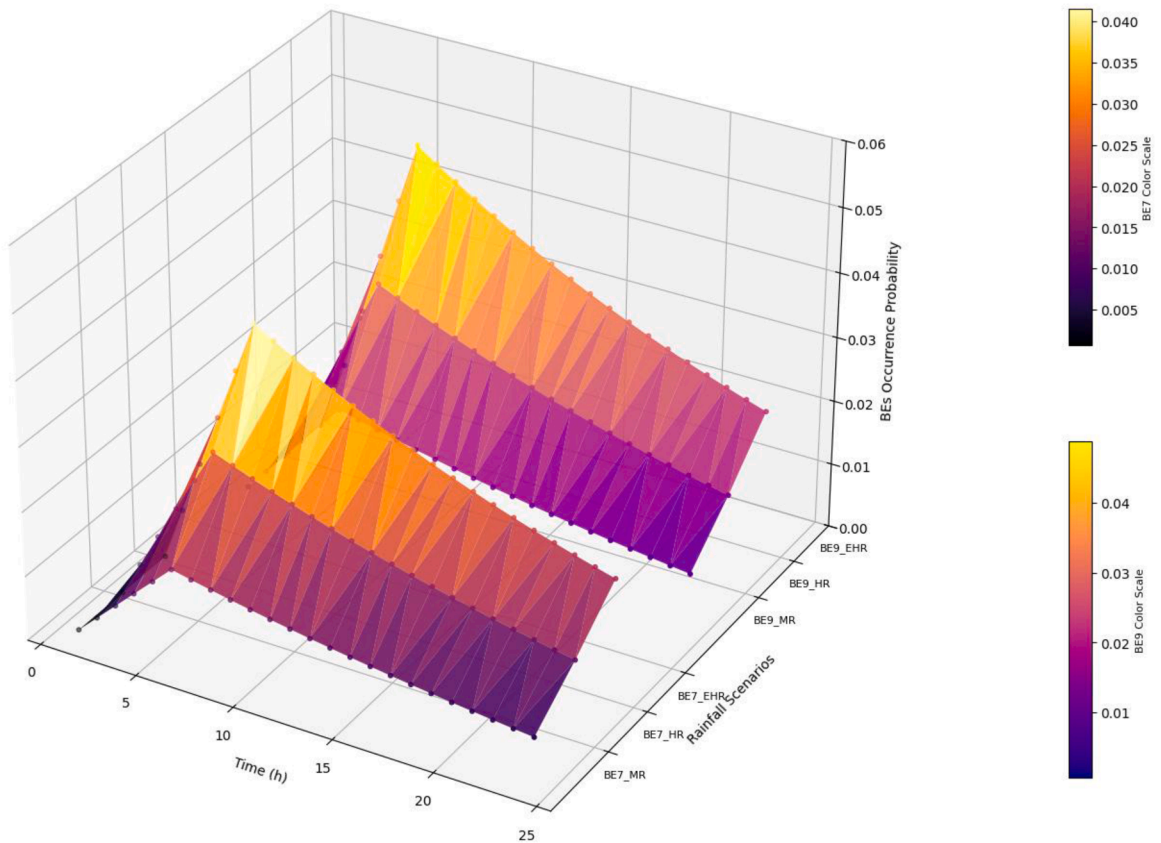


Fig. 14. The occurrence probability of BE6 and BE7 of the electricity distribution system under three rainfall scenarios with recovery strategy 1.

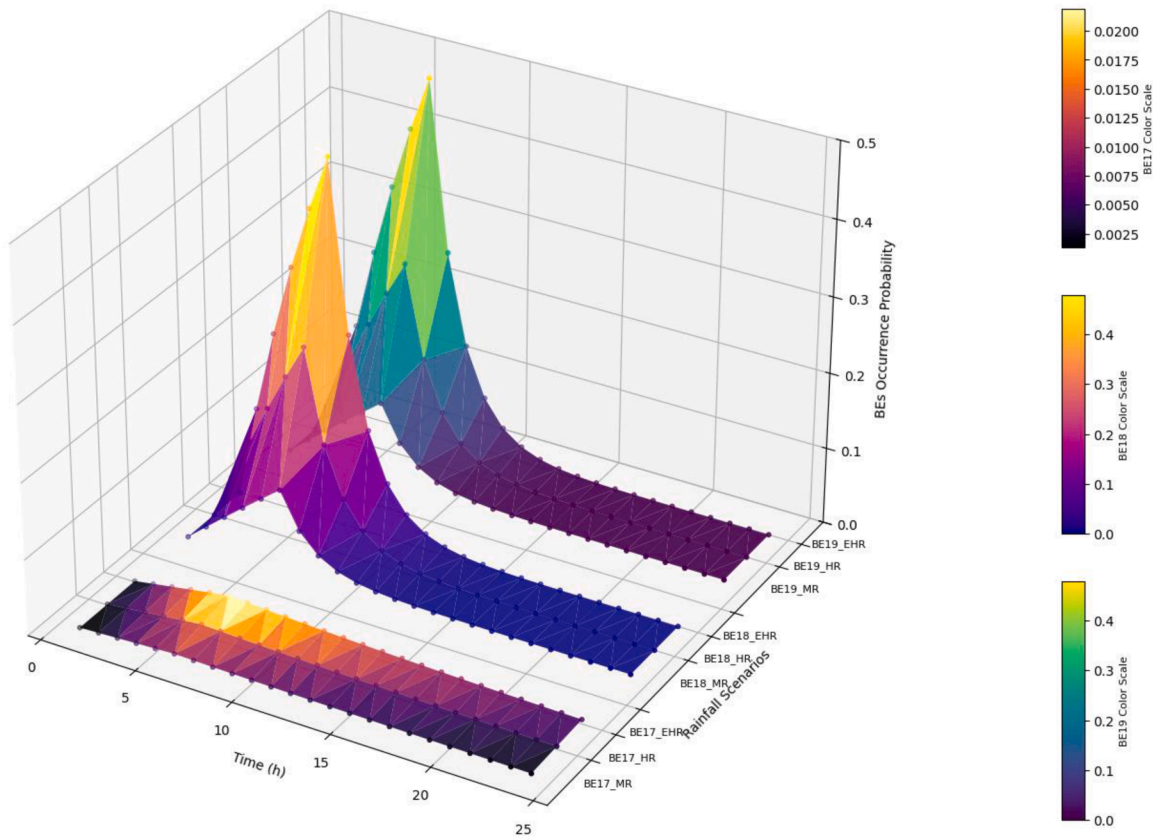


Fig. 15. The occurrence probability of BE17, BE18, and BE19 of the road network under three rainfall scenarios with recovery strategy 1.

moderate rainfall, emphasising the need for scenario-specific resilience planning. Among the three systems, the road network emerges as the most vulnerable due to its direct exposure to rainfall conditions. However, the road network demonstrates a relatively rapid recovery, as evidenced in Fig. 9, suggesting that while its initial impact is high, its recovery is more rapid than the water and electricity distribution systems. In addition, the water supply system experiences more decline in resilience performance than electricity system. This result shows that the water system's resilience performance is not only affected by direct exposure to rainfall but also from its operational dependency on electricity. The interdependency between infrastructure sectors, such as the correlations of water distribution with power supply, becomes a crucial factor in resilience assessment and underscores the need for integrated recovery planning that accounts for cascading effects across systems.

The analysis of recovery strategies reveals that the sequenced recovery strategy 2 outperforms the general recovery strategy 1. In this case study, prioritising the recovery of the road network, followed by the electricity system, and then the water distribution system, yields better overall system resilience. Further insights emerge from the performance analysis of critical components within each infrastructure system. In the water distribution system, pumps serving buildings 1 are more vulnerable than those for Building 2&3. For the electricity distribution system, submersion of components and high humidity pose a greater threat, highlighting the need for protective infrastructure placement in water logging-prone areas. In the road network, the main impact of rainfall is the direct reduction in accessibility, rather than indirect effects on malfunctioning traffic lights. These findings emphasise the importance of differentiating between component-level failure mechanisms and prioritising targeted protective measures accordingly.

The results yield clear managerial implications for resilience planning in interdependent WER systems. Under non-disaster conditions, ageing sub-pipelines exhibit the highest vulnerability, indicating the need for targeted inspection and renewal actions. Under heavy rainfall,

impacts rise nonlinearly with rainfall intensity, underscoring the importance of hazard intensity-specific planning. Practically, early restoration of road accessibility is particularly beneficial because roads can recover relatively quickly and enable cross-sector repairs. Furthermore, coordinated restoration is essential because water resilience performance is strongly constrained by power supply. At the asset level, managers should prioritise hardening pumps serving Building 1, reducing submersion and humidity exposure for electrical assets in waterlogging-prone areas, and focusing road interventions on drainage improvement and rapid clearance.

## 7. Conclusions

This study develops a probabilistic, time-dependent computational model to evaluate the resilience performance of interdependent infrastructure networks at the component level. The main contribution is an integrated FT-DBN pipeline that unifies static dependency logic with dynamic degradation–recovery processes. Specifically, the FT is used to (i) structure inter-component and interdependency between infrastructures and (ii) quantify non-disaster failure probabilities considering component ageing and fatigue; the DBN then extends this to (iii) simulate cascading failures across interdependent infrastructures over time under hazard-induced stressors and recovery strategies. As an outcome, the proposed computational model produces inferable and comparable probabilistic outputs: component-level failure probabilities and system-level resilience performance trajectories, both obtained through time-iterative simulation, which directly support decision-making on “what to protect” and “what to repair first”. The proposed computational model was applied to a case study involving WER systems in a community in northern China. The results demonstrate that the calculated resilience performance values effectively capture both the topological and functional features of infrastructure components, thereby validating the computational model. Under non-disaster

conditions, the analysis identifies water sub-pipelines as relatively vulnerable due to ageing-related corrosion and mechanical deterioration, resulting in a higher baseline probability of water supply failure (1.89 %) compared with electricity supply failure (0.67 %) and road low accessibility (0.58 %).

Under disaster conditions, the DBN-based resilience performance trajectories show that extremely heavy rainfall causes substantially larger performance loss than moderate or heavy rainfall, confirming the need for scenario-specific resilience planning. Among the three infrastructures, the road network exhibits the most pronounced immediate performance drop because it is directly exposed to rainfall, yet it also shows the fastest recovery potential in the case study. Moreover, the recovery strategy comparison indicates that a sequenced restoration strategy (prioritising road recovery, followed by electricity and then water) yields higher overall system resilience than a general strategy, highlighting that effective recovery must account for cross-sector interdependency rather than treating infrastructures independently. Component-level findings further provide practical insights: (i) pumps serving specific buildings are more failure-prone than others, (ii) electrical components are particularly sensitive to submersion and high humidity, and (iii) rainfall-driven accessibility reduction dominates the road performance loss more than traffic-light malfunction. Together, these results show that both system-level interdependencies and component-level correlations are necessary to explain resilience outcomes and to design targeted interventions.

While the proposed FT-DBN computational model provides an interpretable probabilistic simulation of resilience performance, the current treatment of recovery strategy remains simplified. In this paper, repair is represented through an aggregate recovery rate and pre-defined sequencing policies, which effectively model a “return-to-normal” tendency but do not explicitly capture real-world restoration actions, such as discrete repair tasks, crew dispatch, access constraints, spare-part logistics, budget limits, or precedence relationships among repairs. Future work will strengthen the recovery strategy by integrating the DBN with optimal network restoration and resource-constrained recovery planning [15,49]. One promising direction is to couple the DBN’s time-varying failure and service-loss probabilities with optimisation models that select repair actions and allocate limited resources to maximise expected service restoration or minimise expected loss under

uncertainty. In addition, incorporating spatially explicit constraints (co-location exposure, travel time, and waterlogging-prone areas) would further improve realism and external validity.

## Declarations

None.

## Ethics approval and consent to participate

Not applicable.

## Consent for publication

Not applicable.

## CRediT authorship contribution statement

**Wenjing Tong:** Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Formal analysis, Data curation, Conceptualization. **Hong Xian Li:** Writing – review & editing, Supervision, Formal analysis, Conceptualization. **Farnad Nasirzadeh:** Writing – review & editing, Supervision, Formal analysis. **Yingbo Ji:** Supervision, Resources, Investigation, Conceptualization. **Fuyi Yao:** Writing – review & editing, Resources, Project administration, Funding acquisition, Formal analysis, Conceptualization. **Ruichen Zhou:** Resources, Investigation.

## Declaration of competing interest

The authors declare no competing interests.

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## Appendix

Table A1

**Table A1**  
Notation table.

| Symbol              | Meaning                                                              | Unit   | Notes                                                                              |
|---------------------|----------------------------------------------------------------------|--------|------------------------------------------------------------------------------------|
| $\lambda_{f,eq}$    | Failure rate of equipment under heavy-rain conditions                | 1/time | Equipment such as pumps, transformers, traffic lights; Eq. (1)                     |
| RH                  | Relative humidity                                                    | – or % | Use consistently as percent ( %); Eq. (1)                                          |
| D                   | Submersion water depth                                               | m      | Represents inundation exposure; Eq. (1)                                            |
| k1, k2              | Weighting (calibration) coefficients for humidity and submersion     | –      | Estimated empirically or by calibration; Eq. (1)                                   |
| n, m                | Sensitivity exponents for humidity and submersion                    | –      | Equipment-type and exposure dependent; Eq. (1)                                     |
| $\lambda_{f,roads}$ | Rate of low accessibility (failure) of road segments due to rainfall | 1/time | Road accessibility degradation rate; Eq. (2)                                       |
| RI                  | Rainfall intensity                                                   | mm/h   | Eq. (2)                                                                            |
| k3                  | Empirical coefficient for road-segment vulnerability                 | –      | Calibrated from observations; Eq. (2)                                              |
| a                   | Exponent capturing road sensitivity to rainfall intensity            | –      | Road-type/condition dependent; Eq. (2)                                             |
| f(t)                | Weibull probability density function (PDF) of a basic event          | 1/time | Eq. (3)                                                                            |
| $\beta$             | Weibull shape parameter                                              | –      | $\beta < 1$ early failures; $\beta = 1$ constant rate; $\beta > 1$ ageing; Eq. (3) |
| $\eta$              | Weibull scale parameter                                              | time   | Eq. (3)                                                                            |
| $Pr(BE_i)(t)$       | Occurrence probability of basic event i by time t                    | –      | Weibull CDF for component i; Eq. (4)                                               |
| $BE_i$              | Basic event i in the fault tree                                      | –      | Typically component-level failure; Eqs. (4)–(5)                                    |
| $Pr(MCS_j)(t)$      | Occurrence probability of minimal cut set j by time t                | –      | Product of BE probabilities (independence assumed); Eq. (5)                        |
| MCS <sub>j</sub>    | Minimal cut set j                                                    | –      | Smallest BE combination leading to top event; Eqs. (5)–(6)                         |
| m <sub>j</sub>      | Number of basic events within MCS <sub>j</sub>                       | –      | Use m <sub>j</sub> to avoid overloading notation; Eq. (5)                          |

(continued on next page)

Table A1 (continued)

| Symbol         | Meaning                                                               | Unit   | Notes                                                                |
|----------------|-----------------------------------------------------------------------|--------|----------------------------------------------------------------------|
| K              | Number of minimal cut sets (MCSs)                                     | –      | Recommended for Eq. (6) indexing                                     |
| Pr(TE)(t)      | Occurrence probability of the top event by time t                     | –      | Union of MCSs; Eq. (6)                                               |
| TE             | Top event (system-level failure)                                      | –      | Root of the fault tree; Eq. (6)                                      |
| P(Y X1,...,Xn) | Conditional probability of child node Y given parents X1,...,Xn       | –      | DBN CPT definition; Eqs. (7)–(8)                                     |
| Y              | Child (effect) node in DBN                                            | –      | Output node for noisy logic; Eqs. (7)–(8)                            |
| X <sub>j</sub> | Parent (cause) node j in DBN                                          | –      | Input nodes; Eqs. (7)–(8)                                            |
| p <sub>j</sub> | Influence probability parameter associated with parent X <sub>j</sub> | –      | Used in noisy-OR / noisy-AND CPTs; Eqs. (7)–(8)                      |
| Noisy-OR       | Probabilistic OR gate for CPT construction                            | –      | Defines causal activation under uncertainty; Eq. (7)                 |
| Noisy-AND      | Probabilistic AND gate for CPT construction                           | –      | Defines joint activation under uncertainty; Eq. (8)                  |
| λ              | Hazard-induced failure (degradation) rate for time-dependent nodes    | 1/time | Computed from hazard stressors; Markov transition                    |
| μ              | Recovery (repair) rate for time-dependent nodes                       | 1/time | Typically μ = 1/MTTR; derived from repair time data/expert judgement |
| CPT            | Conditional probability table                                         | –      | Parameterisation step after defining DBN structure                   |

## Data availability

Data will be made available on request.

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